

**INS 204: INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN DEPARTMENT
OF COMPUTER SCIENCES**

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**Analysis and Design of Vehicle Services and
Tracking System for NT
Nigeria Limited
(Vehicle Leasing System)**

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1.0 INTRODUCTION

1.1 Background of the Study

Nigeria's corporate transport and logistics industry demands a high level of operational precision, it's not just about getting from point A to point B, but about ensuring safety, reliability, and responsiveness for every client. NT Nigeria Limited has been a part of this landscape since 1995, building a reputation for comprehensive mobility solutions that range from standard vehicle leasing to complex fleet management and manpower recruitment. With a fleet of over 450 vehicles, from the sturdy Toyota Coaster and Hilux to executive Prado SUVs and Corolla sedans; the company serves a wide variety of clients, including diplomatic missions, educational institutions, and corporate organizations across Nigeria and the ECOWAS region.

Currently, day-to-day coordination feels a bit disconnected because the intake process is fragmented. Whether it's a client walking into the reception or sending an inquiry via WhatsApp, the data, customer details, vehicle preferences, and service dates is often captured manually on paper forms or in digital messages. Because these booking methods don't talk to each other, managing the fleet has become an uphill battle. Availability checks often involve tedious manual cross-referencing, while trip monitoring and billing depend on an endless cycle of phone calls and spreadsheet updates.

To move past these bottlenecks, we realized there was an urgent need to modernize. By applying the standard Waterfall System Development Life Cycle (SDLC) framework, this study takes a deep dive into NT Nigeria Limited's existing workflows. Our goal was simple: to design a centralized "Vehicle Services and Tracking System" that brings reservations, driver allocation, fleet tracking, maintenance, and invoicing into one cohesive, enterprise-grade platform.

1.2 Statement of Problem

The current reliance on semi-manual procedures at NT Nigeria Limited creates significant friction in the vehicle leasing process. Our analysis identified five key operational pain points that hinder growth:

1. **Fragmented Intake:** Trying to track reservations across both physical paper forms and scattered

WhatsApp threads is a recipe for errors. We're losing historical data and creating unnecessary administrative work during peak hours.

2. **Scheduling Bottlenecks:** When dispatchers rely on manual registers to verify vehicle availability, the risk of double booking is high. Without a real-time engine to track active leases and workshop schedules, emergency reassignment often becomes a chaotic, last-minute scramble.
3. **Blind Spots in Trip Monitoring:** Even though the vehicles have GPS, the system is detached from the booking software. Dispatchers are still forced to call drivers manually to check on trip progress, route status, or arrival times.
4. **Billing Delays:** Reconciling the bottom line, calculating excess mileage, rental overruns, and overtime requires staff to manually cross-reference paper waybills. This creates significant delays in sending out invoices to corporate accounts.
5. **Reactive Maintenance:** Because maintenance alerts rely on manual odometer logging, it's far too easy to miss service intervals, leading to avoidable mechanical failures while a vehicle is out on a lease.

1.3 Objectives

The goal is to be able to move NT Nigeria Limited away from fragmented processes and into a modern, unified web-based system. By building this "Vehicle Services and Tracking System," we're aiming to tie reservations, fleet monitoring, maintenance, and billing into one single platform.

Specifically, we are looking to achieve the following:

- **Centralized Data:** We're designing a database architecture that keeps all our records(client profiles, contracts, driver schedules, and maintenance history) in one place so we can actually trust the data we're looking at.
- **Smarter Reservations:** We want to build a dispatch module that syncs vehicle availability with location, maintenance status, and active leases, effectively ending the era of "oops, that vehicle is double-booked."
- **Live Tracking:** Instead of checking external dashboards, we're integrating telematics data right into the

core app. This lets us monitor where our vehicles are, check their live odometer readings, and see trip progress instantly.

- **Proactive Maintenance:** We're moving from reactive fixes to a preventive model. The system will now automatically flag service alerts based on real-time mileage or inspection dates, stopping breakdowns before they happen.
- **Billing Accuracy:** Our new billing module will automate the boring stuff. It'll handle lease rates, calculate excess mileage, and track driver overtime, so when we send an invoice, it's itemized, accurate, and ready for the client immediately.
- **Client Transparency:** We're adding a client-facing portal, giving our corporate partners a secure way to log in, view their leased vehicles, check on their trips, and download their own transaction summaries.

1.4 Project's Scope and Constraints

The focus of this project is to streamline the core operations of NT Nigeria Limited's vehicle hire and leasing department:

- **What's In:** We're covering everything from digital onboarding (for walk-ins and referrals) to generating contracts and rate cards. We're also tackling the full dispatch flow; allocating drivers, automating waybills, and managing check-ins/check-outs. On the fleet side, we're integrating location tracking and mileage-driven maintenance schedules. Finally, we're automating the billing engine for corporate invoices.
- **What's Out (Constraints):** Because this is a 16-week academic project, we're focusing on the SAD and Waterfall development phases. We aren't manufacturing custom GPS hardware; instead, we're simulating those data feeds through standard APIs. Also, we're sticking to open-source web technologies and cloud infrastructure to keep the recommended deployment cost-effective and compliant with Nigeria Data Protection Act (NDPA) standards.

1.5 Alternative Solutions

We looked at a few different ways to tackle the current operational challenges, and here is how they stacked up:

Alternative	Advantages	Disadvantages / Bottlenecks
Option A: Retain & Enhance Disconnected Software Tools (Enhance Current Tools)	• Lowest upfront capital expenditure. • Minimal staff technical training required. • Immediate continuation of current reception and WhatsApp workflows.	• Fails to eliminate booking schedule clashes across vehicle classes. • Does not synchronize live GPS telematics with reservation calendars. • Continued manual delays in calculating invoices.
Option B: Off-the-Shelf ERP Software	• Fast deployment with pre-built templates. • Mature commercial codebase.	• Prohibitive licensing fees in foreign currency. • Rigid workflows that do not accommodate local structures. • High customization expenses.
Option C: Custom In-House Web-Based System <i>[Selected]</i>	• Built specifically around NT's leasing workflows. • 100% software ownership. • Integrates all reservation and tracking feeds.	• Demands upfront design and development effort. • Requires structured staff onboarding.

1.6 System Description

We're designing the "NT VMS" as a three-tier web application.

- **The Front-End (Presentation Tier):** This is the user-friendly interface for our receptionists, dispatchers, mechanics, and accountants. It needs to be responsive and intuitive so everyone can do their job without getting bogged down by complexity.
- **The Logic (Application Tier):** This is the "brain" of the operation. It handles the business rules, which is verifying if a car is free, assigning the right driver, ingesting GPS data, triggering maintenance alerts, and running the complex math for invoices.
- **The Data (Data Tier):** We're using a PostgreSQL relational database. It ensures our data stays consistent, secure, and most importantly backed up, so we never lose a booking record again.

1.7 Feasibility Assessment

1.7.1 Economic Analysis (Cost-Benefit Analysis)

Economic: We ran the numbers over a three-year cycle. While there's an initial investment in hardware and development, the ROI is clear: we stop losing money on unbilled excess mileage, eliminate the administrative cost of paper-based logging, and prevent expensive, avoidable vehicle breakdowns.

Technical: This is highly feasible. Our current office network and broadband are robust enough to run this web platform, and the technologies we've chosen (Node.js/Express, HTML5) are standard industry tools that our development team is well-prepared to implement.

Schedule: We are sticking to a 16-week Waterfall roadmap, moving sequentially through planning, analysis, design, coding, and testing to ensure we hit our deadline.

Cost Category	Year 0 (Setup - NGN)	Year 1 (NGN)	Year 2 (NGN)	Year 3 (NGN)
Hardware Procurement	₦2,650,000	-	-	-

Software & Cloud Hosting	₦250,000	₦120,000	₦130,000	₦140,000
Custom System Development	₦850,000	-	-	-
Staff Training & Management	₦350,000	₦60,000	₦60,000	₦60,000
Maintenance & Support	-	₦240,000	₦260,000	₦280,000
TOTAL OPERATIONAL COSTS	₦4,100,000	₦420,000	₦450,000	₦480,000

1.7.2 Technical Analysis

The technical assessment evaluated the availability of computing hardware, networking connectivity, software technologies, and internal technical capability at NT Nigeria Limited. The company's administrative headquarters and base operations yard maintain stable broadband internet access and an internal Local Area Network connecting the reception, operations, workshop, and accounting offices. The software architecture leverages reliable, industry-standard open-source technologies, including a PostgreSQL relational database management system, a Node.js/Express backend service layer, and a responsive HTML5/CSS3/JavaScript front-end. The technical parameters and telematics API ingestion scripts are fully within the technical execution capability of the systems development team, confirming that the technical risks are completely mitigated.

1.7.3 Schedules, Timeline, and Resource Analysis

The project execution follows the Waterfall System Development Life Cycle (SDLC) structured across a strict 16-week timeline:

SDLC Phase	Duration	Key Activities	Roles
1: Planning	Weeks 1 – 3	Preliminary investigation, stakeholder interviews, cost-benefit study.	Leader, Logistics Analyst

2: Analysis	Weeks 4 – 7	Process modeling (DFDs), data modeling (ERDs), SRS creation.	Senior Analyst, Secretary
3: Design	Weeks 8 – 11	Architectural design, normalization, screen layouts, module specs.	Software & Data Designers
4: Coding	Weeks 12 – 14	Database deployment, API development, telematics ingestion.	Software & Data Designers
5: Testing	Weeks 15 – 16	Unit, integration, security, and UAT, onboarding and handover.	Full Project Team

1.8 Recommendation

After weighing our options, the team strongly recommends developing the **Custom In-House Web-Based System (NT VMS)**. It is the only solution that integrates our multi-channel booking fragmentation into a single calendar, connects live telematics to our dispatch process, and automates our billing cycle. It transforms the fleet from a set of disconnected assets into a data-driven service operation.

2.0 ANALYSIS OF THE EXISTING SYSTEM

2.1 Process Description of the System (using Context and DFDs)

We took a close look at how NT Nigeria Limited currently handles vehicle bookings, and frankly, the process is a bit fragmented. Right now, requests come in from three different channels: walk-ins, phone referrals, and their WhatsApp Business line.

Here is what we observed happens behind the scenes:

- 1. The Intake Chaos:** When a customer walks into reception, they're handed a paper slip. If they reach out via WhatsApp, they get a digital form. It sounds simple, but because these records don't live in the same place, the data isn't unified.

2. **The "Availability" Guessing Game:** Before confirming anything, the receptionist has to flip through physical ledgers and scroll through shared spreadsheets. It's manual, slow, and prone to human error.
3. **The Dispatch Scramble:** Once a booking is confirmed, assigning a car and driver is a manual process. It requires issuing paper waybills and, often, a lot of back-and-forth phone calls to see who is actually available.
4. **The Monitoring Gap:** Even though the vehicles have GPS trackers, that system is totally separate from the booking software. Dispatchers have to switch between tabs and call drivers just to get basic status updates.
5. **The End-of-Trip Audit:** When a car returns, staff have to manually inspect it, log the odometer, and verify fuel levels on paper. It's a bottleneck that delays everything that follows.

2.1.1 Current Physical Data Flow Diagram (CPDFD)

The diagram below illustrates the physical flow of paper documents, ledger entries, and human movement in the current vehicle leasing workflow:

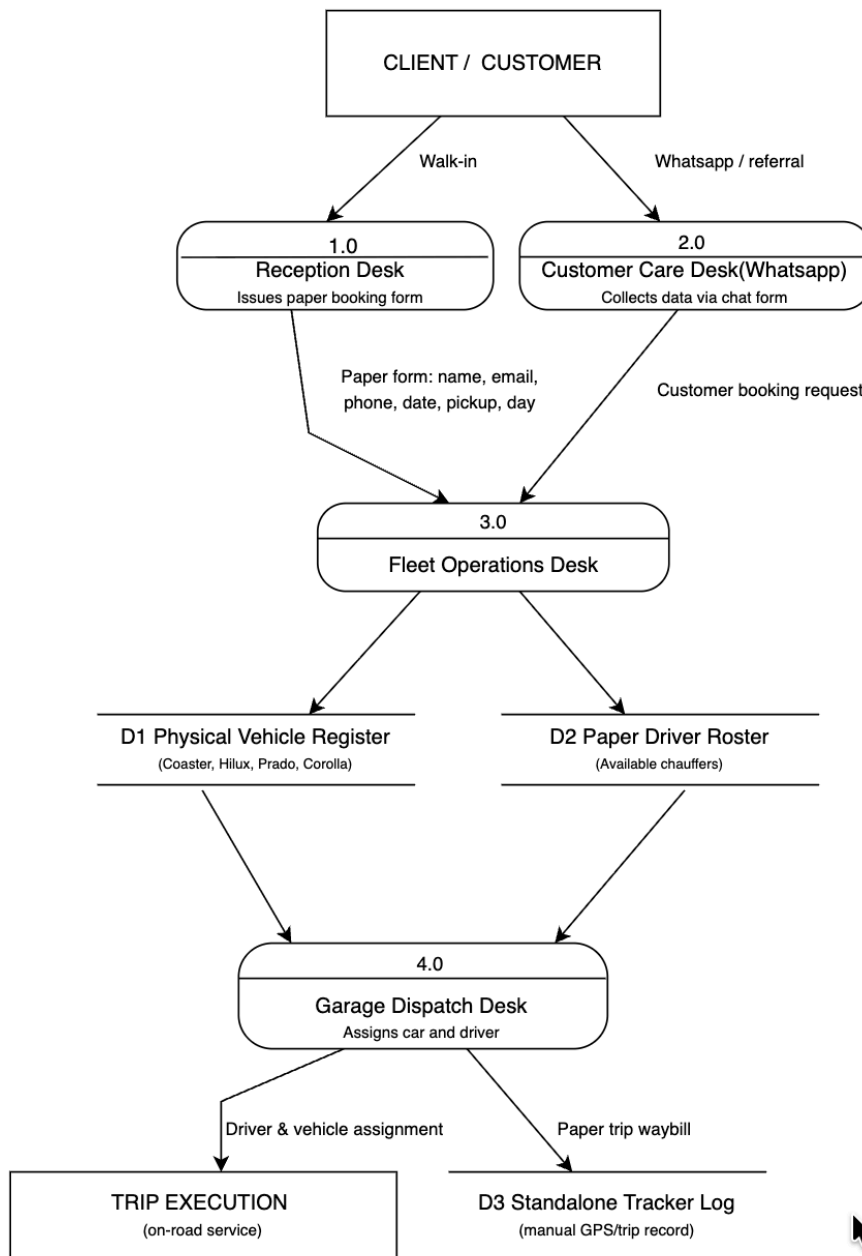


Figure 2.1: Current Physical Data Flow Diagram (CPDFD) of Vehicle Leasing Workflow

2.1.2 Current Logical Data Flow Diagram

The logical view of the existing system abstracts away physical media (paper cards, walking) to reflect purely the essential data transformations occurring in the current operational environment.

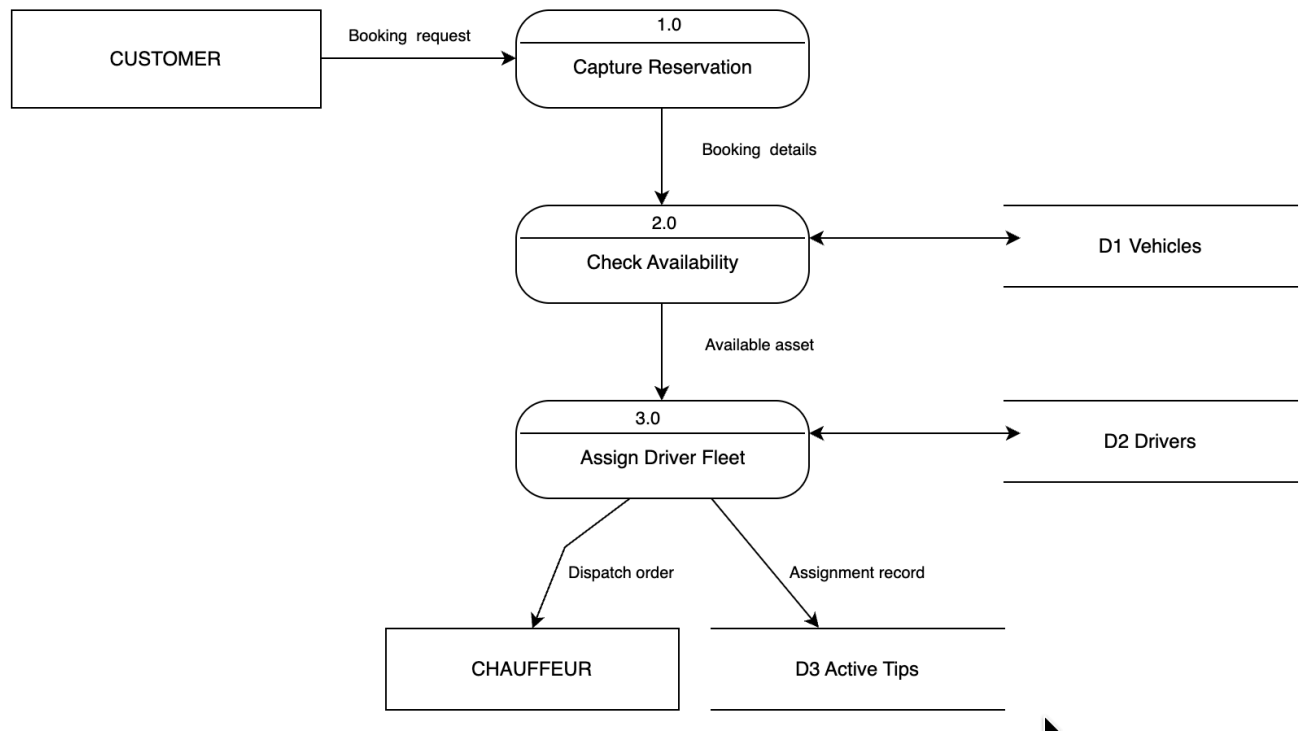


Figure 2.2: Current Logical Data Flow Diagram (Level-0)

2.2 Data Dictionary of Existing System

To get a handle on the data, we cataloged the specific pieces of information that the team currently juggles. Seeing it laid out like this really highlights why the current manual approach creates so many blind spots.

Data Structure / Flow	Description & Data Elements	Origin & Destination
Customer_Booking_Data	Customer_Name + Email + Phone_Number + Service_Date + Pickup_Point + Duration_Days + Preferred_Category	Customer → Reception / Whatsapp Desk
Vehicle_Inventory_Entry	Vehicle_Reg_No + Make_Model (Coaster, Hilux, Prado, Corolla)	Operations Desk ↔ Vehicle Register Ledger

	+ Chassis_No + Current_Status (Available, On-Lease, In-Workshop)	
Driver_Roster_Entry	Driver_ID + Driver_Name + Phone + License_No + Availability_Status (On-Duty, Idle, On-Leave)	Operations ↔ Driver Roster Ledger
Trip_Waybill_Slip	Waybill_No + Customer_Name + Vehicle_Reg_No + Driver_Name + Start_Date + Return_Date + Initial_Odometer	Dispatch Desk ↔ Chauffeur / Active Trip File

2.3 Data Description of the System (using ERDs)

When we mapped out the relationships between the entities (Customers, Bookings, Vehicles, and Drivers), we found that the existing "manual domain" has very loose connections. Because everything is paper-based, there's no automated integrity check. For example, a driver could be assigned to a trip without the system inherently knowing their license is expired, because that data isn't linked to the dispatch record.

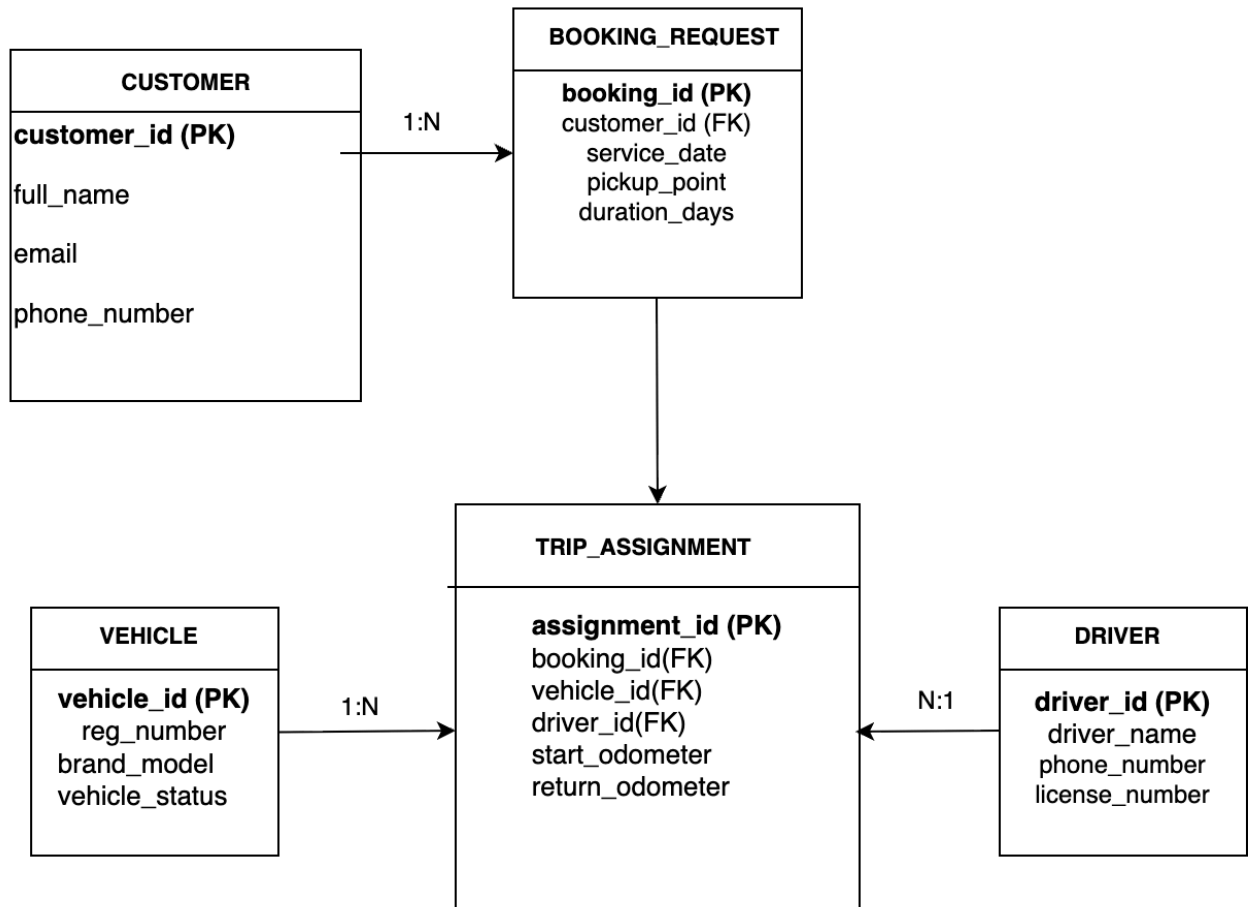


Figure 2.3: Entity-Relationship Diagram of Existing Manual Domain

2.4 Problem Areas

After spending time with the operations team, we identified five major pain points that we absolutely had to address:

1. **Data Fragmentation:** Because we're balancing paper logs and WhatsApp chats, we effectively have two different truths. This makes it nearly impossible to keep a clean customer history.
2. **Scheduling Clashes:** Without a real-time availability engine, the risk of "oops, that car is already booked" is constant, especially during high-demand periods.
3. **Disconnected Tracking:** GPS data shouldn't be a separate silo. Dispatchers are wasting time acting as human relays between the tracker and the customer.
4. **Billing Latency:** The manual effort required to reconcile paper waybills is the main reason invoices are

going out late. It's a drag on cash flow.

5. **Reactive Maintenance:** We're relying on memory or manual logbooks to track service intervals. This is a huge risk as it turns a simple oil change into a breakdown on the highway.

3.0 DESIGN OF THE PROPOSED SYSTEM

3.1 System Overview

Our vision for the **NT Vehicle Management & Tracking System (NT VMS)** isn't just about putting a digital skin on old habits; it's about creating a living, breathing command center. We're moving away from the "desk-to-desk" paper trail and building an integrated hub where every department from the first hello at reception to the final payment entry in accounting, is looking at the exact same data in real-time.

By ditching manual ledgers for role-based automation and syncing live GPS feeds directly into our dispatch logic, we're essentially giving the operations team superpowers. No more guessing where a Hilux is or whether a driver is due for a break; the system handles the heavy lifting so the team can focus on service quality rather than logistics puzzles.

3.2 Process Specification

3.2.1 Required Logical Data Flow Diagram

To make sure we didn't miss a single step in the transition, we mapped out the new logical flow. We wanted to clearly define how data should move between customers and our internal teams without getting tripped up by "physical" barriers like paper waybills or having to walk to another office to check a register.

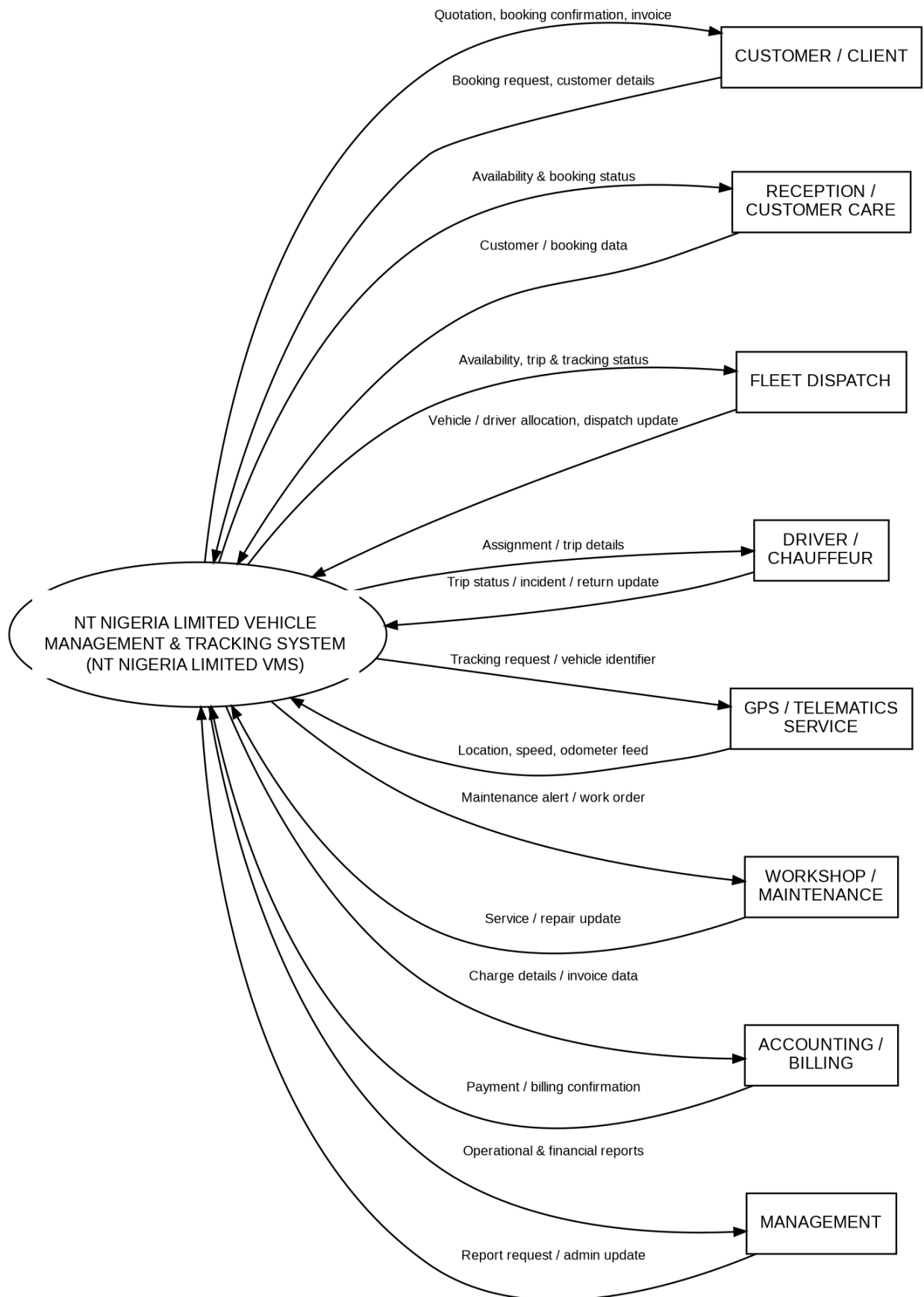


Figure 3.1: Context Diagram (Level 0) for Proposed NT Nigeria Limited VMS

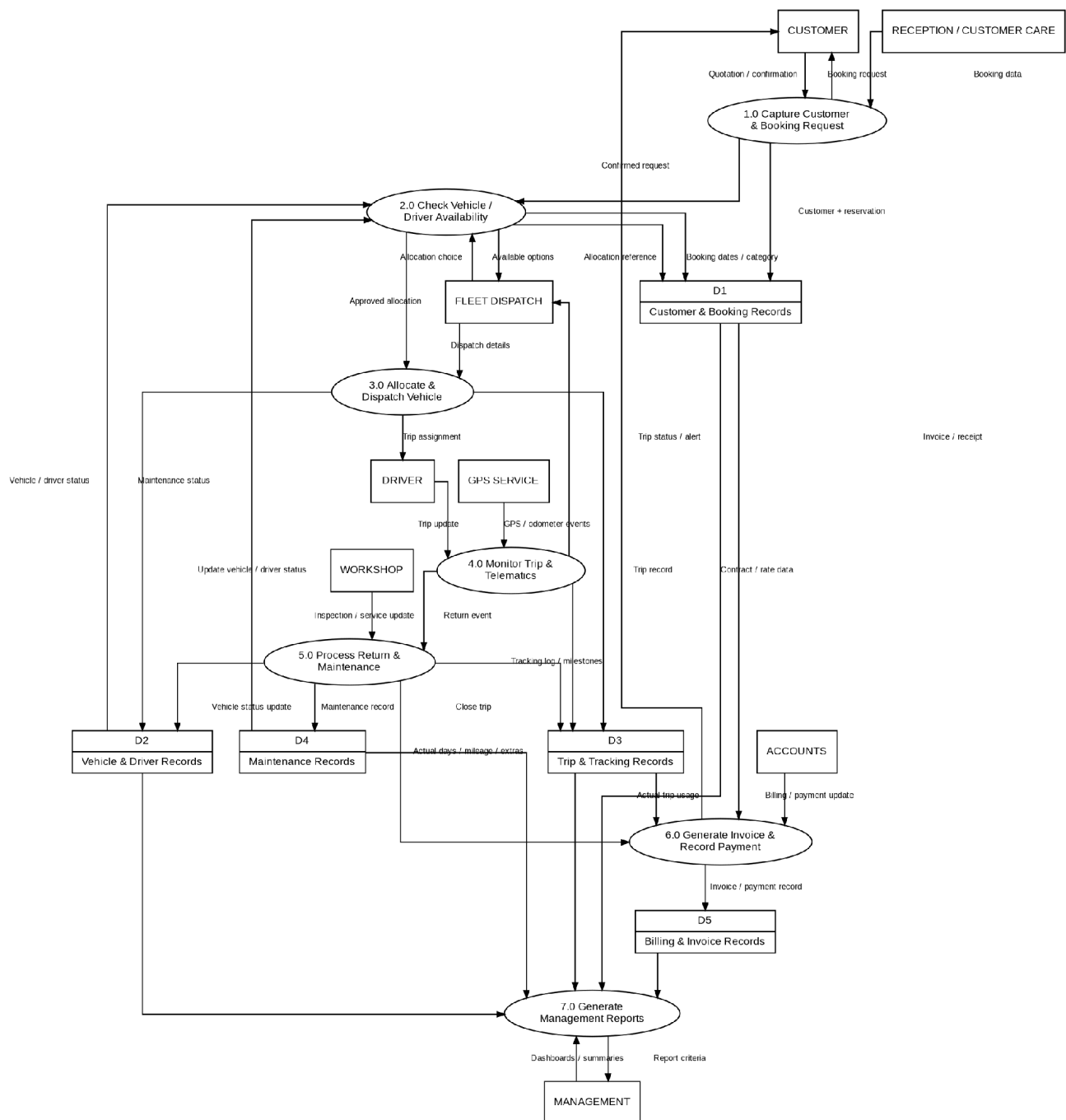


Figure 3.2: Required Logical Data Flow Diagram (Level 1)

The key shift in our new DFD is the introduction of five "shared memories" our digital data stores. By centralizing everything from customer contracts to live telematics in these stores, we've effectively connected the dots between booking, monitoring, and billing. This means when a trip ends, the billing engine already knows the mileage from the tracking log without anyone having to type it in.

3.2.2 Data Dictionary (Proposed System)

Think of this dictionary as the vocabulary of our new system. We've redefined these elements to ensure that every department uses the same terms, eliminating the confusion that happens when returned to a driver means something different than returned to an accountant.

Data Flows

Data Flow	Description	From	To	Data Elements
Customer Booking Request	Customer service request for vehicle hire or lease.	Customer / Reception	P1.0	Customer Name, Phone, Email, Service Date, Return Date, Pickup Location, Vehicle Category, Service Type
Booking Details Entered	Reservation details captured by receptionist or customer care officer.	Reception / Customer Care	P1.0	Customer ID, Vehicle Category, Start Date, Return Date, Pickup Location, Duration
Quotation / Booking Confirmation	Price and reservation status returned to customer after booking review.	P1.0	Customer	Reservation ID, Estimated Amount, Status, Terms
Booking Record Saved	Confirmed or pending booking stored for further processing.	P1.0	D1	Reservation ID, Customer ID, Dates, Vehicle Category, Booking Status
Availability Request	Request to identify a suitable vehicle and chauffeur for the booking period.	P1.0	P2.0	Reservation ID, Vehicle Category, Start Date, Return Date, Driver Required
Vehicle Availability Data	Current vehicle category and operational status	D2	P2.0	Vehicle ID, Registration No, Category, Status, Current Mileage

	used during allocation.			
Maintenance Status	Maintenance state checked before a vehicle can be allocated.	D4	P2.0	Vehicle ID, Maintenance Status, Due Date, Due Mileage
Driver Availability Data	Current chauffeur status and licence details used during assignment.	D2	P2.0	Driver ID, Driver Name, License Expiry, Availability Status
Allocation Details	Approved vehicle and driver selected for a reservation.	P2.0	P3.0 / D1	Reservation ID, Vehicle ID, Driver ID, Allocation Status
Dispatch Details	Vehicle release information recorded before trip begins.	Fleet Dispatch	P3.0	Dispatch ID, Vehicle ID, Driver ID, Initial Odometer, Dispatch Time, Fuel Level
Driver Assignment	Trip instructions transmitted to the assigned chauffeur.	P3.0	Driver	Trip ID, Customer, Pickup Point, Service Date, Vehicle Details
Trip Record Saved	Active trip record created when a vehicle is dispatched.	P3.0	D3	Trip ID, Dispatch ID, Start Time, Trip Status
GPS / Telematics Feed	Live vehicle location, speed and odometer information.	GPS Service	P4.0	Vehicle ID, Latitude, Longitude, Speed, Odometer, Event Time
Trip Status / Alert	Active trip status and exceptions sent to fleet dispatch.	P4.0	Fleet Dispatch	Trip ID, Vehicle Location, Last Update, Alert Type
Return Details	Vehicle return information submitted at end of trip.	Driver / Fleet Dispatch	P5.0	Trip ID, Return Time, Final Odometer, Fuel Level, Condition Notes
Maintenance Work Order	Maintenance instruction created	P5.0	Workshop / D4	Maintenance ID, Vehicle ID, Service

	from mileage, due date or vehicle fault.			Type, Trigger, Priority, Status
Service Update	Workshop completion or repair status returned to system.	Workshop	P5.0	Maintenance ID, Work Done, Cost, Completion Date, Next Service Due
Billing Input	Actual trip data and contract information passed for charge calculation.	D1 / D3	P6.0	Reservation ID, Base Rate, Actual Days, Actual Mileage, Driver Charge, Penalty
Invoice Record	Itemized lease invoice generated and stored.	P6.0	D5 / Customer	Invoice ID, Reservation ID, Charge Breakdown, Total Amount, Due Date, Status
Payment Update	Payment details recorded against invoice.	Accounts	P6.0 / D5	Payment ID, Invoice ID, Amount Paid, Method, Reference, Status
Report Request	Management request for operational or financial summary.	Management	P7.0	Report Type, Date From, Date To, Filters
Management Report	Generated report delivered to management.	P7.0	Management	Fleet Status, Utilization, Maintenance Due, Invoice Summary, Date Range

Data Stores

Data Store	ID	Description	Data Elements
Customer & Booking Records	D1	Stores customer profiles, reservations, rate references and lease/contract details.	Customer ID (PK), Reservation ID (PK), Rate ID (FK), Service Dates, Vehicle Category, Booking Status, Contract No

Vehicle & Driver Records	D2	Stores all fleet vehicles and chauffeurs used for availability and dispatch decisions.	Vehicle ID (PK), Registration No, Make, Model, Category, Mileage, Vehicle Status, Driver ID (PK), License No, Driver Status
Trip & Tracking Records	D3	Stores dispatch, active trip, return and GPS/telematics event records.	Dispatch ID (PK), Trip ID (PK), Vehicle ID (FK), Driver ID (FK), Start/End Odometer, Location, Speed, Event Time, Trip Status
Maintenance Records	D4	Stores preventive and corrective maintenance activities and vehicle workshop status.	Maintenance ID (PK), Vehicle ID (FK), Service Type, Trigger Mileage, Due Date, Fault Description, Status, Cost
Billing & Invoice Records	D5	Stores invoices and payments linked to completed or billable reservations.	Invoice ID (PK), Reservation ID (FK), Charge Breakdown, Total Amount, Invoice Status, Payment ID (PK), Payment Reference, Payment Status

Data Elements

Data Element	Description	Type	Example
Customer ID	Unique identifier for each client or organization.	Numeric	1001
Reservation ID	Unique identifier for each booking.	Text / Numeric	RES-2026-0184
Vehicle ID	Unique identifier for each fleet vehicle.	Numeric	V0452
Registration Number	Official vehicle registration number.	Text	LSD-842XY
Vehicle Category	Vehicle class requested or assigned.	Text	Hilux / Prado / Corolla / Coaster
Booking Status	Current stage of the reservation.	Text	Pending / Confirmed / Cancelled / Completed
Driver ID	Unique identifier for each chauffeur.	Numeric	D118

Driver Status	Availability condition of a chauffeur.	Text	Available / Assigned / On Leave
Current Mileage	Most recent accepted vehicle odometer reading.	Numeric	87,450 km
Trip ID	Unique identifier for an active/completed trip.	Text	TRP-00918
Latitude / Longitude	GPS coordinate of vehicle at an event time.	Decimal	6.5244, 3.3792
Maintenance Status	Current workshop/service state.	Text	Due / In Workshop / Completed
Base Lease Rate	Standard approved rate for the lease period.	Currency (₦)	₦95,000/day
Excess Mileage Rate	Charge applied above approved mileage allowance.	Currency (₦)	₦450/km
Invoice Status	Current stage of generated invoice.	Text	Draft / Issued / Paid / Overdue
Payment Method	Method used by customer to settle invoice.	Text	Transfer / Card / Cash
Payment Reference	Reference used to identify a payment.	Text	NTN-TRX-20260827-44

3.2.3 Decision Tables

To take the guesswork out of daily operations, we have translated NT Nigeria's business expertise into these decision tables. This ensures the system acts as a second set of eyes for our dispatchers and workshop managers, enforcing safety and policy rules automatically.

Decision Table 1: Vehicle and Driver Allocation

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
CONDITIONS					
Booking Confirmed?	Y	Y	Y	Y	N

Requested Vehicle Category Available?	Y	Y	N	Y	-
Selected Vehicle Free from Maintenance Hold?	Y	N	-	Y	-
Driver Required?	N	-	-	Y	-
Eligible Driver Available?	-	-	-	Y	-
ACTIONS					
Confirm Vehicle Allocation	✓			✓	
Assign Driver	-	-	-	✓	-
Reject Selected Vehicle		✓			
Suggest Alternative Vehicle / Date		✓	✓		
Return Booking for Confirmation					✓

Y = Condition true | N = Condition false | - = Irrelevant | ✓ = Action performed

Decision Table 2: Maintenance Trigger and Vehicle Release

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
CONDITIONS				
Return Inspection Completed?	Y	Y	Y	N
Critical Fault / Damage Detected?	N	Y	N	-
Service Mileage / Due Date Reached?	N	-	Y	-

ACTIONS				
Mark Vehicle Available	✓			
Create Maintenance Work Order		✓	✓	
Mark Vehicle In Workshop / Hold		✓	✓	✓
Send Maintenance Alert		✓	✓	✓

3.3 Data Specification

3.3.1 Entity Relationship Diagram (ERD) – Proposed System

Our new data model is designed to fix the loose connection we found in the manual system. By creating hard links between vehicles, maintenance logs, and dispatch records, we've made it impossible for a vehicle with an open fault to be accidentally assigned to a new trip. It's built-in accountability that doesn't rely on someone remembering to check a clipboard.

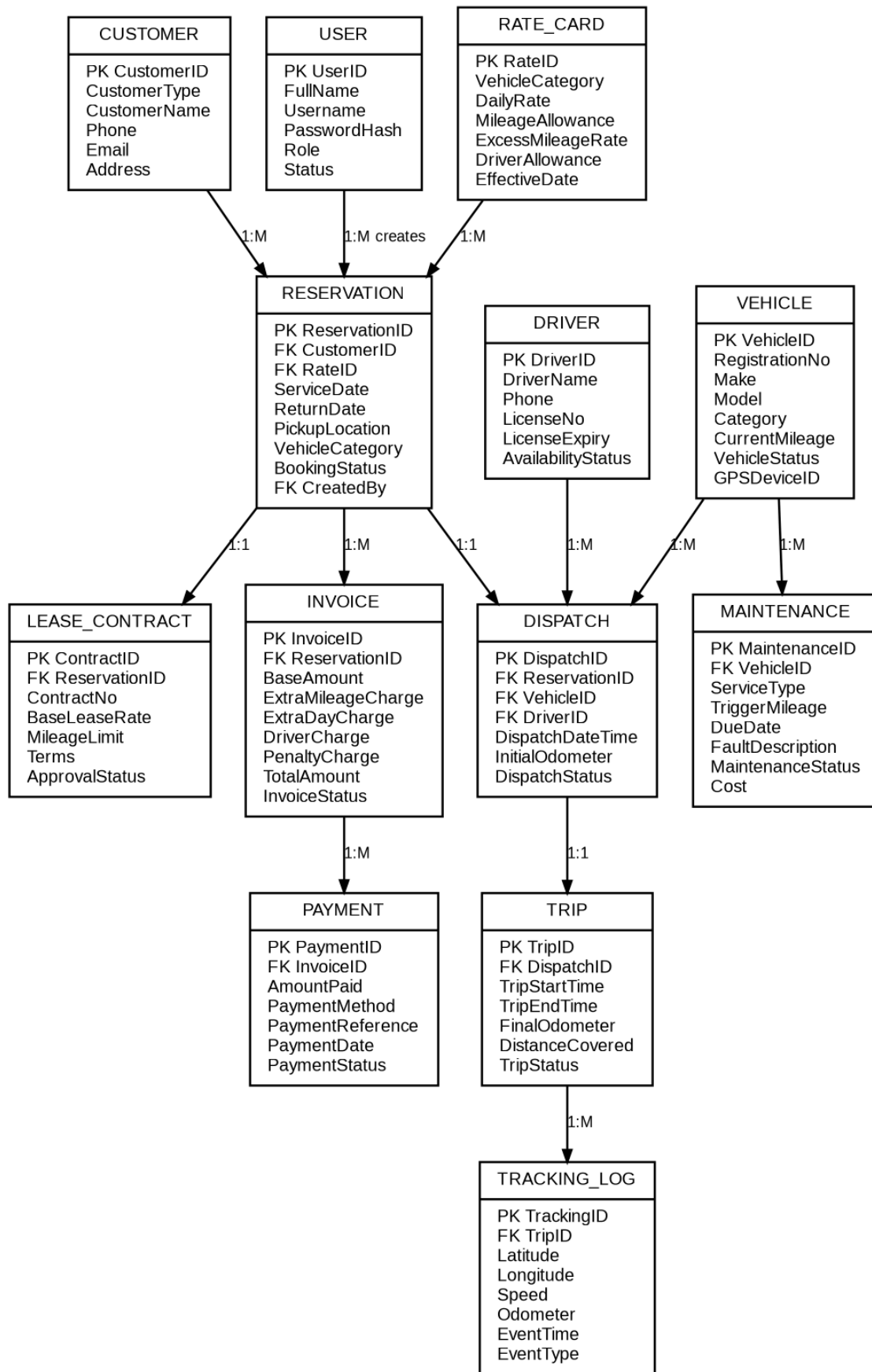


Figure 3.3: Entity Relationship Diagram – Proposed NT Nigeria Limited VMS

3.3.2 Database Design (Tables from ERDs)

The following relational tables are derived from the proposed ERD:

Table 1: USER

Field Name	Data Type	Size	Constraint	Description
UserID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique identifier for each system user
FullName	VARCHAR	120	NOT NULL	Full name of staff user
Username	VARCHAR	80	UNIQUE, NOT NULL	Login username or email
PasswordHash	VARCHAR	255	NOT NULL	Encrypted/hashed password value
Role	VARCHAR	40	NOT NULL	Reception, Dispatch, Workshop, Accounts, Management/Admi n
UserStatus	VARCHAR	20	DEFAULT Active	Active or Inactive account

Table 2: CUSTOMER

Field Name	Data Type	Size	Constraint	Description
CustomerID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique identifier for customer/client
CustomerType	VARCHAR	30	NOT NULL	Individual, Corporate, Embassy, Institution
CustomerName	VARCHAR	150	NOT NULL	Customer or organization name
PhoneNumber	VARCHAR	20	NOT NULL	Primary customer phone number
Email	VARCHAR	120	NULL	Customer email address

Address	VARCHAR	250	NULL	Customer or organization address
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Table 3: VEHICLE

Field Name	Data Type	Size	Constraint	Description
VehicleID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique identifier for fleet vehicle
RegistrationNo	VARCHAR	20	UNIQUE, NOT NULL	Vehicle registration number
Make	VARCHAR	50	NOT NULL	Vehicle manufacturer
Model	VARCHAR	80	NOT NULL	Vehicle model
Category	VARCHAR	40	NOT NULL	Coaster, Hilux, Prado, Corolla, etc.
CurrentMileage	DECIMAL	(10,1)	DEFAULT 0	Latest accepted odometer reading
VehicleStatus	VARCHAR	30	NOT NULL	Available, Reserved, On Lease, In Workshop, Out of Service
GPSDeviceID	VARCHAR	80	NULL	Identifier used by tracking platform

Table 4: DRIVER

Field Name	Data Type	Size	Constraint	Description
DriverID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique identifier for chauffeur
DriverName	VARCHAR	120	NOT NULL	Full name of driver/chauffeur
PhoneNumber	VARCHAR	20	NOT NULL	Driver contact number
LicenseNo	VARCHAR	50	UNIQUE, NOT NULL	Driver licence number
LicenseExpiry	DATE	--	NOT NULL	Driver licence expiry date
AvailabilityStatus	VARCHAR	30	NOT NULL	Available, Assigned, On Leave, Inactive

Table 5: RATE CARD

Field Name	Data Type	Size	Constraint	Description
------------	-----------	------	------------	-------------

RateID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique pricing record
VehicleCategory	VARCHAR	40	NOT NULL	Vehicle class to which rate applies
DailyRate	DECIMAL	(12,2)	NOT NULL	Base daily hire/lease rate
MileageAllowance	DECIMAL	(10,1)	NULL	Included mileage where applicable
ExcessMileageRate	DECIMAL	(12,2)	DEFAULT 0	Charge for excess mileage
DriverAllowance	DECIMAL	(12,2)	DEFAULT 0	Approved chauffeur allowance/rate
EffectiveDate	DATE	--	NOT NULL	Date from which rate is active

Table 6: RESERVATION

Field Name	Data Type	Size	Constraint	Description
ReservationID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique booking identifier
CustomerID	INT	--	FOREIGN KEY → CUSTOMER(Cu stomerID)	Customer making the booking
RateID	INT	--	FOREIGN KEY → RATE_CARD(Ra teID)	Rate applied to booking
CreatedBy	INT	--	FOREIGN KEY → USER(UserID)	Staff member who created booking
ServiceDate	DATETIME	--	NOT NULL	Requested start date/time
ReturnDate	DATETIME	--	NOT NULL	Expected end/return date/time
PickupLocation	VARCHAR	250	NOT NULL	Customer pickup/service location
VehicleCategory	VARCHAR	40	NOT NULL	Requested vehicle class
BookingStatus	VARCHAR	30	DEFAULT Pending	Pending, Quoted, Confirmed, Cancelled, Completed

Table 7: LEASE CONTRACT

Field Name	Data Type	Size	Constraint	Description
ContractID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique lease/contract record
ReservationID	INT	--	FOREIGN KEY → RESERVATION(ReservationID)	Reservation covered by contract
ContractNo	VARCHAR	60	UNIQUE, NOT NULL	Official lease/booking contract number
BaseLeaseRate	DECIMAL	(12,2)	NOT NULL	Approved base lease amount/rate
MileageLimit	DECIMAL	(10,1)	NULL	Approved mileage allowance/limit
Terms	TEXT	--	NULL	Contract terms and special conditions
ApprovalStatus	VARCHAR	30	DEFAULT Pending	Pending, Approved, Rejected

Table 8: DISPATCH

Field Name	Data Type	Size	Constraint	Description
DispatchID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique dispatch record
ReservationID	INT	--	FOREIGN KEY → RESERVATION(ReservationID)	Reservation being dispatched
VehicleID	INT	--	FOREIGN KEY → VEHICLE(VehicleID)	Vehicle allocated
DriverID	INT	--	FOREIGN KEY → DRIVER(DriverID), NULLABLE	Assigned chauffeur where required
DispatchDateTime	DATETIME	--	NOT NULL	Actual vehicle release time
InitialOdometer	DECIMAL	(10,1)	NOT NULL	Odometer reading at dispatch
DispatchStatus	VARCHAR	30	DEFAULT Released	Prepared, Released, Cancelled

Table 9: TRIP

Field Name	Data Type	Size	Constraint	Description
------------	-----------	------	------------	-------------

TripID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique trip identifier
DispatchID	INT	--	FOREIGN KEY → DISPATCH(DispatchID)	Dispatch linked to trip
TripStartTime	DATETIME	--	NOT NULL	Actual trip start time
TripEndTime	DATETIME	--	NULL	Actual trip return/completion time
FinalOdometer	DECIMAL	(10,1)	NULL	Odometer reading on return
DistanceCovered	DECIMAL	(10,1)	NULL	Calculated distance travelled
TripStatus	VARCHAR	30	DEFAULT Active	Active, Completed, Delayed, Incident

Table 10: TRACKING LOG

Field Name	Data Type	Size	Constraint	Description
TrackingID	BIGINT	--	PRIMARY KEY, AUTO INCREMENT	Unique telematics event record
TripID	INT	--	FOREIGN KEY → TRIP(TripID)	Active trip receiving the event
Latitude	DECIMAL	(10,7)	NOT NULL	Vehicle latitude
Longitude	DECIMAL	(10,7)	NOT NULL	Vehicle longitude
Speed	DECIMAL	(7,2)	NULL	Recorded vehicle speed
Odometer	DECIMAL	(10,1)	NULL	Telematics odometer reading
EventTime	DATETIME	--	NOT NULL	Time event was received/recorded
EventType	VARCHAR	40	NULL	Location, Overspeed, Geofence, Offline, etc.

Table 11: MAINTENANCE

Field Name	Data Type	Size	Constraint	Description
MaintenanceID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique maintenance/work order
VehicleID	INT	--	FOREIGN KEY →	Vehicle being maintained

			VEHICLE(VehicleID)	
ServiceType	VARCHAR	100	NOT NULL	Routine service, repair, inspection, etc.
TriggerMileage	DECIMAL	(10,1)	NULL	Mileage that triggered service
DueDate	DATE	--	NULL	Scheduled maintenance due date
FaultDescription	TEXT	--	NULL	Reported defect or inspection finding
MaintenanceStatus	VARCHAR	30	DEFAULT Open	Open, In Progress, Completed
Cost	DECIMAL	(12,2)	DEFAULT 0	Recorded maintenance cost

Table 12: INVOICE

Field Name	Data Type	Size	Constraint	Description
InvoiceID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique invoice record
ReservationID	INT	--	FOREIGN KEY → RESERVATION(ReservationID)	Booking being billed
BaseAmount	DECIMAL	(12,2)	NOT NULL	Approved base lease charge
ExtraMileageCharge	DECIMAL	(12,2)	DEFAULT 0	Excess mileage charge
ExtraDayCharge	DECIMAL	(12,2)	DEFAULT 0	Charge for additional rental days
DriverCharge	DECIMAL	(12,2)	DEFAULT 0	Driver allowance/overtime charge
PenaltyCharge	DECIMAL	(12,2)	DEFAULT 0	Late-return or approved penalty
TotalAmount	DECIMAL	(12,2)	NOT NULL	Total invoice value
InvoiceStatus	VARCHAR	30	DEFAULT Draft	Draft, Issued, Paid, Overdue, Cancelled

Table 13: PAYMENT

Field Name	Data Type	Size	Constraint	Description
------------	-----------	------	------------	-------------

PaymentID	INT	--	PRIMARY KEY, AUTO INCREMENT	Unique payment record
InvoiceID	INT	--	FOREIGN KEY → INVOICE(InvoiceID)	Invoice being settled
AmountPaid	DECIMAL	(12,2)	NOT NULL	Amount paid by customer
PaymentMethod	VARCHAR	30	NOT NULL	Transfer, Card, Cash, etc.
PaymentReference	VARCHAR	100	NULL	Bank/transaction reference
PaymentDate	DATETIME	--	NOT NULL	Date and time of payment
PaymentStatus	VARCHAR	30	DEFAULT Confirmed	Pending, Confirmed, Reversed

Relationship Summary

Relationship	Type	Foreign Key
CUSTOMER → RESERVATION	One to Many	RESERVATION.CustomerID
USER → RESERVATION	One to Many	RESERVATION.CreatedBy
RATE CARD → RESERVATION	One to Many	RESERVATION.RateID
RESERVATION → LEASE CONTRACT	One to One	LEASE_CONTRACT.ReservationID
RESERVATION → DISPATCH	One to One	DISPATCH.ReservationID
VEHICLE → DISPATCH	One to Many	DISPATCH.VehicleID
DRIVER → DISPATCH	One to Many	DISPATCH.DriverID
DISPATCH → TRIP	One to One	TRIP.DispatchID
TRIP → TRACKING LOG	One to Many	TRACKING_LOG.TripID
VEHICLE → MAINTENANCE	One to Many	MAINTENANCE.VehicleID
RESERVATION → INVOICE	One to Many	INVOICE.ReservationID

INVOICE → PAYMENT	One to Many	PAYMENT.InvoiceID
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3.4 Screen Layout / Specification

We have designed these screens with the busy dispatcher in mind. The goal was to reduce the number of clicks and browser tabs someone needs to manage, putting the most critical info like live locations and urgent maintenance alerts; front and center.

1. LOGIN SCREEN

NT Nigeria Limited VMS

Login

NT NIGERIA LIMITED

Vehicle Management & Tracking System

Username / Email

Password

Role

2. CUSTOMER BOOKING & RESERVATION SCREEN

NT Nigeria Limited VMS

Customer Booking & Reservation

CUSTOMER INFORMATION

Customer / Organisation

Phone Number

Email Address

Pickup Location

SERVICE REQUEST

Vehicle Category

Start Date

Return Date

Service Type

Special Requirements

Rate / Contract

Estimated Amount

Booking Status: DRAFT

SAVE DRAFT

GENERATE QUOTE

CONFIRM BOOKING

CANCEL

3. FLEET AVAILABILITY & DISPATCH SCREEN

NT Nigeria Limited VMS

Fleet Availability & Dispatch

Booking: RES-2026-0184 | Customer: ABC Energy Ltd | Category: Hilux | 27-29 Aug 2026

AVAILABLE VEHICLES

Reg No	Make/Model	Status	Odometer	Maint.	
LSD-842XY	Toyota Hilux	Available	87,450 km	OK	☐
KJA-420TR	Toyota Hilux	Available	64,120 km	Due in 1,800 km	☒
AAA-115PK	Toyota Hilux	On Lease	92,060 km	OK	☐

DRIVER / CHAUFFEUR

Name: Emeka Okafor

Status: Available

Licence Expiry: 18/02/2027

Current Assignment: None

SELECT DRIVER

Dispatch Details

Initial Odometer

Fuel Level

Condition / Release Note

CONFIRM ALLOCATION

CREATE WAYBILL

DISPATCH

4. LIVE VEHICLE TRACKING SCREEN

NT Nigeria Limited VMS

Live Fleet Tracking

1

2

3

MAP / LIVE VEHICLE LOCATION VIEW

ACTIVE TRIP

Trip ID

TRP-00918

Vehicle

LSD-842XY - Hilux

Driver

Emeka Okafor

Customer

ABC Energy Ltd

Last Update

07:41:22

Speed

62 km/h

Odometer

87,633 km

Trip Status

IN TRANSIT

Alert: No active exception

VIEW TRIP

CONTACT DRIVER

5. MAINTENANCE MANAGEMENT SCREEN

NT Nigeria Limited VMS
Workshop Maintenance

MAINTENANCE QUEUE

Vehicle	Trigger	Service / Fault	Due	Status	Action
KJA-420TR	Mileage	Routine 70,000 km service	28/08/26	Due Soon	Open
APP-044AZ	Inspection	Front brake vibration	Immediate	Hold	Open
LSR-777KD	Date	Roadworthiness inspection	02/09/26	Scheduled	View

WORK ORDER DETAILS

Vehicle

Service Type

Fault / Work Description

Current Odometer

Cost

Next Service Odometer

Status

SAVE WORK ORDER

MARK COMPLETE

6. BILLING & INVOICE SCREEN

NT Nigeria Limited VMS
Corporate Billing & Invoice

Reservation: RES-2026-0184 | Customer: ABC Energy Ltd | Contract: LC-0184

Charge Item	Basis	Rate	Qty	Amount
Base Vehicle Lease	Daily lease	₦95,000	3 days	₦285,000
Excess Mileage	Mileage above allowance	₦450/km	120 km	₦54,000
Driver Allowance	Chauffeur service	₦18,000/day	3 days	₦54,000
Late Return Penalty	Approved rule	₦15,000	1	₦15,000
				TOTAL: ₦408,000

Invoice Status
Payment Status
Payment Reference

Draft

Unpaid

Optional

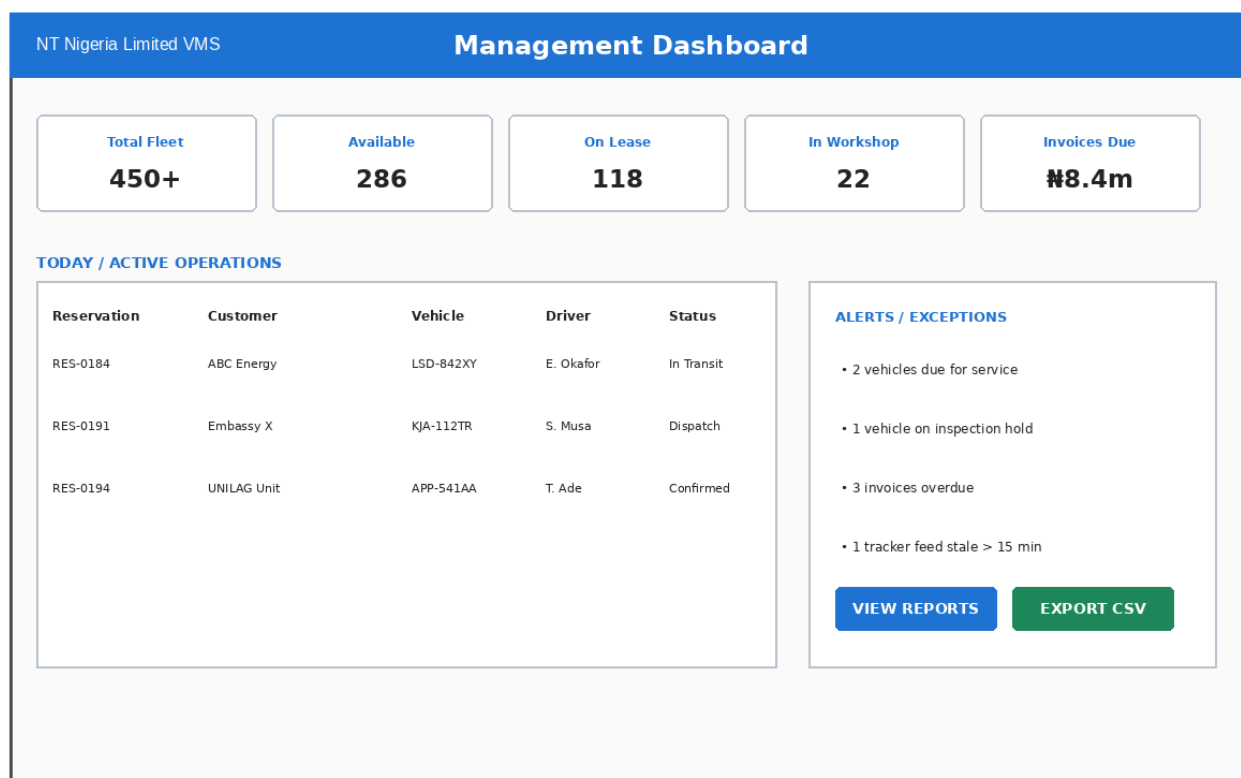
SAVE DRAFT

ISSUE INVOICE

RECORD PAYMENT

PRINT PDF

7. MANAGEMENT DASHBOARD



3.5 Report / Form Specifications

Even in a digital system, documents still matter. We have designed these reports and forms to provide the clarity management has been missing, turning raw trip data into itemized invoices and clear utilization snapshots that help us see which parts of the fleet are working the hardest.

Form 1: Customer Booking / Reservation Form (F-001)

Form Name	Customer Booking / Reservation Form
Form Number	F-001
Purpose	To capture customer details, requested service dates, vehicle category, pickup location and booking requirements.
Used By	Reception / Customer Care
Trigger	Opened when a new customer booking request is received.
Data Destination	Customer & Booking Records (D1)
Validation Rules	Customer name and phone are required; Service Date must be before Return Date; Vehicle Category must be selected; a confirmed booking must have a valid rate/contract reference.

NT Nigeria Limited VMS
Customer Booking & Reservation

CUSTOMER INFORMATION

Customer / Organisation

Phone Number

Email Address

Pickup Location

SERVICE REQUEST

Vehicle Category

Start Date

Return Date

Service Type

Special Requirements

Rate / Contract

Estimated Amount

Booking Status: DRAFT

SAVE DRAFT

GENERATE QUOTE

CONFIRM BOOKING

CANCEL

CUSTOMER BOOKING / RESERVATION FORM LAYOUT

Form 2: Vehicle Dispatch / Trip Waybill Form (F-002)

Form Name	Vehicle Dispatch / Trip Waybill Form
Form Number	F-002
Purpose	To record the assigned vehicle and driver and authorize release for a confirmed reservation.
Used By	Fleet Dispatch
Trigger	Opened after a confirmed reservation passes vehicle/driver availability checks.
Data Destination	Vehicle & Driver Records (D2), Trip & Tracking Records (D3)
Validation Rules	Vehicle must be available and not under maintenance hold; driver must be eligible where required; Initial Odometer must be entered before dispatch.

NT Nigeria Limited VMS
Fleet Availability & Dispatch

Booking: RES-2026-0184 | Customer: ABC Energy Ltd | Category: Hilux | 27-29 Aug 2026

AVAILABLE VEHICLES

Reg No	Make/Model	Status	Odometer	Maint.	
LSD-842XY	Toyota Hilux	Available	87,450 km	OK	☐
KJA-420TR	Toyota Hilux	Available	64,120 km	Due in 1,800 km	☒
AAA-115PK	Toyota Hilux	On Lease	92,060 km	OK	☐

DRIVER / CHAUFFEUR

Name: Emeka Okafor
Status: Available
Licence Expiry: 18/02/2027
Current Assignment: None

SELECT DRIVER

Dispatch Details

Initial Odometer
Fuel Level
Condition / Release Note

CONFIRM ALLOCATION
CREATE WAYBILL
DISPATCH

VEHICLE DISPATCH / TRIP WAYBILL FORM LAYOUT

Form 3: Return Inspection & Maintenance Form (F-003)

Form Name	Return Inspection & Maintenance Form
Form Number	F-003
Purpose	To capture final odometer, vehicle condition, faults and maintenance triggers when a trip is completed.
Used By	Fleet Dispatch / Workshop
Trigger	Opened when the assigned vehicle returns from a trip.
Data Destination	Trip & Tracking Records (D3), Maintenance Records (D4), Vehicle Records (D2)
Validation Rules	Final Odometer must not be lower than Initial Odometer; critical faults must place vehicle on hold; maintenance due mileage/date must be checked before vehicle is marked Available.

NT Nigeria Limited VMS
Workshop Maintenance

MAINTENANCE QUEUE

Vehicle	Trigger	Service / Fault	Due	Status	Action
KJA-420TR	Mileage	Routine 70,000 km service	28/08/26	Due Soon	Open
APP-044AZ	Inspection	Front brake vibration	Immediate	Hold	Open
LSR-777KD	Date	Roadworthiness inspection	02/09/26	Scheduled	View

WORK ORDER DETAILS

Vehicle

Service Type

Fault / Work Description

Current Odometer

Cost

Next Service Odometer

Status

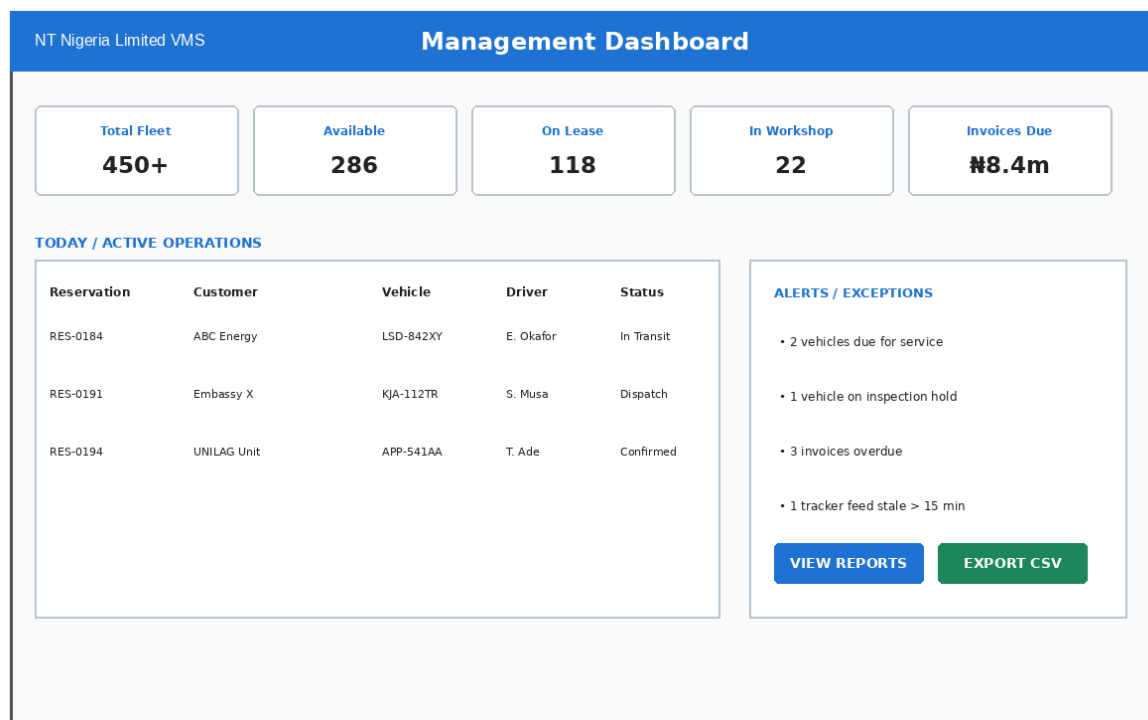
SAVE WORK ORDER

MARK COMPLETE

RETURN INSPECTION / MAINTENANCE FORM LAYOUT

Report 1: Fleet Status & Utilization Report (R-001)

Report Name	Fleet Status & Utilization Report
Report Number	R-001
Purpose	To give management a summary of available, reserved, on-lease, in-workshop and out-of-service vehicles and fleet utilization within a selected period.
Generated By	Management / Fleet Operations
Trigger	On demand by selected date range and vehicle category.
Data Source	Vehicle & Driver Records (D2), Customer & Booking Records (D1), Trip & Tracking Records (D3)
Output Format	On-screen dashboard, printable PDF and exportable CSV
Frequency	On demand – daily, weekly or monthly



FLEET STATUS & UTILIZATION REPORT LAYOUT

Report 2: Maintenance Due & Workshop Report (R-002)

Report Name	Maintenance Due & Workshop Report
Report Number	R-002
Purpose	To display vehicles due for service by date or mileage, vehicles currently in workshop, open faults and completed maintenance records.
Generated By	Workshop / Management
Trigger	On demand or automatically when maintenance thresholds are reached.
Data Source	Maintenance Records (D4), Vehicle Records (D2)
Output Format	On-screen display, printable PDF and exportable CSV
Frequency	Daily / weekly / on demand

Report 3: Billing & Invoice Summary Report (R-003)

Report Name	Billing & Invoice Summary Report
Report Number	R-003
Purpose	To summarize invoices, billed amounts, paid invoices, outstanding balances and payment methods for a selected date range or customer.
Generated By	Accounts / Management
Trigger	On demand by date range, customer or invoice status.
Data Source	Billing & Invoice Records (D5), Customer & Booking Records (D1)
Output Format	On-screen display, printable PDF and exportable CSV

Frequency	Weekly, monthly or on demand
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NT Nigeria Limited VMS

Corporate Billing & Invoice

Reservation: RES-2026-0184 | Customer: ABC Energy Ltd | Contract: LC-0184

Charge Item	Basis	Rate	Qty	Amount
Base Vehicle Lease	Daily lease	₦95,000	3 days	₦285,000
Excess Mileage	Mileage above allowance	₦450/km	120 km	₦54,000
Driver Allowance	Chauffeur service	₦18,000/day	3 days	₦54,000
Late Return Penalty	Approved rule	₦15,000	1	₦15,000
				TOTAL: ₦408,000

Invoice Status

Payment Status

Payment Reference

SAVE DRAFT

ISSUE INVOICE

RECORD PAYMENT

PRINT PDF

BILLING & INVOICE REPORT LAYOUT

Report 4: Customer Lease Invoice (R-004)

Report Name	Customer Lease Invoice
Report Number	R-004
Purpose	To provide an itemized customer invoice showing base lease amount and approved additional charges such as extra mileage, extra days, driver allowance/overtime or penalties.
Generated By	System / Accounts
Trigger	Generated when a billable trip is completed or when Accounts finalizes the billing record.
Data Source	Customer & Booking Records (D1), Trip & Tracking Records (D3), Billing & Invoice Records (D5)
Output Format	PDF / printable invoice
Frequency	Once per invoice or approved billing cycle

3.6 Program / Module Specifications

We broke the system down into these specific modules to tackle each friction point we identified earlier. Each module is a specialized tool designed to solve a real-world headache, like automated billing replacing manual math or live tracking replacing constant check-in calls.

Module 1: User Authentication Module

Module Name	UserAuthentication
Purpose	Validates user credentials and grants role-based access to the NT Nigeria Limited VMS.
Used By	All users (Reception, Dispatch, Workshop, Accounts, Management)
Input	Username, Password, Role
Output	Access granted or denied; user redirected to appropriate dashboard.
Process Logic	System receives login credentials, checks stored user record and role, starts a session for an active user and records the login event.

Function Name	Description
validateLogin()	Checks username and password against stored credentials.
assignRole()	Determines the dashboard and permissions available to the user.
logoutUser()	Ends the user session and returns to the login screen.

Module 2: Customer & Reservation Module

Module Name	ReservationManagement
Purpose	Handles customer registration, quotation and booking/reservation processing.
Used By	Reception / Customer Care
Input	Customer details, Service Date, Return Date, Vehicle Category, Pickup Location, Rate/Contract
Output	Customer and reservation records saved to D1; quotation/booking status returned.
Process Logic	Reception searches or creates the customer profile, captures service requirements, applies an approved rate or contract reference, validates dates and saves the reservation.

Function Name	Description
searchCustomer()	Finds an existing customer or corporate account.
createCustomer()	Creates a new customer record.
createReservation()	Saves a new vehicle service request.
generateQuotation()	Calculates or retrieves the applicable estimated charge.
updateBookingStatus()	Changes booking status after confirmation, cancellation or completion.

Module 3: Fleet Allocation & Dispatch Module

Module Name	FleetDispatch
Purpose	Checks vehicle/driver availability, assigns resources and releases vehicles for confirmed bookings.
Used By	Fleet Dispatch
Input	Reservation ID, Vehicle Category, Vehicle ID, Driver ID, Initial Odometer
Output	Allocation and dispatch records saved to D1/D2/D3; trip assignment sent to driver.
Process Logic	System checks date overlap, vehicle operational status and maintenance records, checks driver availability where required, saves the allocation and creates the dispatch/trip record.

Function Name	Description
findAvailableVehicles()	Returns vehicles in the requested category that are free for the booking period.
checkMaintenanceStatus()	Blocks vehicles that are in workshop or under maintenance hold.
findAvailableDrivers()	Returns eligible chauffeurs without conflicting assignments.
assignVehicleDriver()	Links selected vehicle and driver to the reservation.
createDispatch()	Records check-out details and creates the active trip.

Module 4: Fleet Tracking & Telematics Module

Module Name	FleetTracking
Purpose	Monitors active trips using GPS and odometer information linked to the dispatched vehicle.
Used By	Fleet Dispatch / Management
Input	Vehicle ID, Trip ID, GPS Location, Speed, Odometer, Event Time
Output	Tracking events saved to D3; current trip position and alerts displayed.
Process Logic	System receives telematics data, links the GPS device to the active trip, updates last-known location and odometer values and flags configured tracking exceptions.

Function Name	Description
ingestGPSData()	Receives location and telematics records from the tracking service.

mapVehicleToTrip()	Links the tracked vehicle/device to its current trip.
updateTripLocation()	Stores latest coordinates and odometer reading.
detectTripAlert()	Generates configured delay, geofence, overspeed or stale-feed alerts.

Module 5: Maintenance Management Module

Module Name	MaintenanceManagement
Purpose	Schedules and records preventive and corrective maintenance and controls vehicle workshop availability.
Used By	Workshop / Fleet Dispatch
Input	Vehicle ID, Odometer, Fault Details, Service Type, Due Date, Status
Output	Maintenance record saved to D4; vehicle status updated in D2.
Process Logic	System evaluates return inspection, mileage and service dates, creates maintenance work orders when required and keeps the vehicle unavailable until workshop completion and release.

Function Name	Description
checkServiceDue()	Checks mileage/date thresholds for preventive maintenance.
createWorkOrder()	Creates a new maintenance or repair record.
updateMaintenanceStatus()	Records workshop progress and completion.
updateNextService()	Stores the next due date or odometer threshold.
releaseVehicle()	Returns a completed and cleared vehicle to Available status.

Module 6: Billing & Invoicing Module

Module Name	BillingInvoicing
Purpose	Calculates lease charges and records invoices and customer payments.
Used By	Accounts / Billing
Input	Reservation ID, Contract/Rate, Actual Days, Actual Mileage, Driver Charges, Penalties
Output	Invoice/payment records saved to D5; invoice produced for customer.
Process Logic	System retrieves approved contract/rate details and actual trip usage, computes billable items, generates an itemized invoice and records payment details when received.

Function Name	Description
----------------------	--------------------

loadBillingData()	Retrieves reservation, contract and completed trip details.
calculateLeaseCharges()	Calculates base rate and applicable extra charges.
generateInvoice()	Creates and stores the itemized customer invoice.
recordPayment()	Saves payment method, amount and reference.
updateInvoiceStatus()	Updates Draft, Issued, Paid or Overdue status.

Module 7: Reporting & Administration Module

Module Name	ReportingAdministration
Purpose	Generates operational and financial reports and allows approved master-data administration.
Used By	Management / Admin
Input	Date Range, Vehicle Category, Customer, Report Type, Admin Updates
Output	Dashboards, PDF/CSV reports and updated configuration records.
Process Logic	Management selects report criteria, the system queries the relevant data stores, computes summary totals and displays/export reports. Authorized administrators can also maintain users, vehicle categories and approved rate records.

Function Name	Description
generateFleetReport()	Produces vehicle status and utilization summaries.
generateMaintenanceReport()	Produces maintenance due and workshop summaries.
generateBillingReport()	Produces invoice/payment summaries.
exportReport()	Exports report output to PDF/CSV.
manageSystemUsers()	Creates, edits or disables staff accounts and roles.

3.7 Test Plans

We aren't just testing code; we're testing our operational logic. These plans are designed to break the system before it reaches the yard, ensuring it handles every curveball like a driver with an expired license or a last-minute vehicle breakdown exactly how it's supposed to.

Test Plan 1: User Authentication (TC-001)

Objective	Verify that the system correctly grants or denies access based on credentials and user role.
------------------	--

Test Case	Input	Expected Output	Status
-----------	-------	-----------------	--------

Valid login	Username: fleet01; correct password; Role: Dispatch	Login successful; Fleet Dispatch dashboard opens.	Not Yet Executed
Incorrect password	Valid username; wrong password	Error: Invalid credentials; access denied.	Not Yet Executed
Inactive user	Valid credentials for inactive account	Error: Account inactive; access denied.	Not Yet Executed
Unauthorized module access	Reception user opens Billing module	Access denied based on role permission.	Not Yet Executed

Test Plan 2: Customer Booking & Reservation (TC-002)

Objective	Verify that customer requests and reservations are correctly captured, validated and saved.
------------------	---

Test Case	Input	Expected Output	Status
Create valid booking	Customer: ABC Energy; Hilux; 27/08/2026 to 29/08/2026; Pickup: Victoria Island	Reservation saved to D1 with generated Reservation ID and Pending/Quoted status.	Not Yet Executed
Return date before service date	Start: 29/08/2026; Return: 27/08/2026	Error: Return Date must be later than Service Date.	Not Yet Executed
No vehicle category selected	Required fields completed; Category blank	Error: Vehicle Category is required.	Not Yet Executed
Confirm booking	Valid quotation/contract reference and customer approval	Booking status changes to Confirmed and becomes available for allocation.	Not Yet Executed

Test Plan 3: Vehicle & Driver Allocation (TC-003)

Objective	Verify that the system prevents conflicting or invalid allocations and assigns suitable resources.
------------------	--

Test Case	Input	Expected Output	Status
Allocate available vehicle	Confirmed Hilux booking; vehicle LSD-842XY Available; no date conflict	Vehicle allocated; reservation linked to vehicle.	Not Yet Executed
Vehicle already booked	Selected vehicle overlaps another confirmed reservation	Allocation rejected; alternative vehicle/date suggested.	Not Yet Executed

Vehicle in workshop	Selected vehicle status: In Workshop	Allocation blocked even if no reservation overlap.	Not Yet Executed
Driver licence expired	Driver licence expiry before service date	Driver cannot be assigned; eligible alternatives shown.	Not Yet Executed
Valid vehicle + driver	Both available and eligible	Allocation saved and dispatch can be created.	Not Yet Executed

Test Plan 4: Fleet Tracking & Telematics (TC-004)

Objective	Verify that GPS and trip events are correctly linked to active trips and displayed to fleet operations.
------------------	---

Test Case	Input	Expected Output	Status
Receive valid GPS update	Known GPSDeviceID; active Trip ID; coordinates and odometer supplied	Tracking event saved to D3; map/last-known status updates.	Not Yet Executed
Unknown device	GPS update from unmapped Device ID	Event rejected/flagged for review; no wrong trip update.	Not Yet Executed
Overspeed event	Speed above configured threshold	Trip alert generated and visible on tracking dashboard.	Not Yet Executed
Stale tracking feed	No update beyond configured freshness period	System displays stale/offline tracker warning.	Not Yet Executed

Test Plan 5: Return & Maintenance (TC-005)

Objective	Verify that return inspection, maintenance triggers and vehicle availability are handled correctly.
------------------	---

Test Case	Input	Expected Output	Status
Normal return	Final odometer > initial; no fault; service not due	Trip closed; vehicle returned to Available status.	Not Yet Executed
Invalid odometer	Final odometer lower than initial odometer	Validation error; trip cannot be closed until corrected.	Not Yet Executed
Fault detected	Return inspection records critical brake fault	Maintenance work order created; vehicle status set to In Workshop/Hold.	Not Yet Executed
Service threshold reached	Current mileage crosses next service threshold	Preventive maintenance	Not Yet Executed

		alert/work order created.	
--	--	---------------------------	--

Test Plan 6: Billing & Invoicing (TC-006)

Objective	Verify that lease charges, invoices and payment records are calculated and stored accurately.
------------------	---

Test Case	Input	Expected Output	Status
Base lease only	3-day booking; no excess mileage or penalties	Invoice total equals approved base lease charge.	Not Yet Executed
Excess mileage	Actual mileage exceeds allowance by 120 km; rate ₦450/km	Extra Mileage Charge = ₦54,000 and included in invoice total.	Not Yet Executed
Extra rental day	Actual return is one billable day after contracted return	Extra Day Charge added according to approved rate.	Not Yet Executed
Record payment	Invoice issued; customer pays full amount by transfer with reference	Payment saved to D5; invoice status changes to Paid.	Not Yet Executed

Test Plan 7: Reporting (TC-007)

Objective	Verify that management reports are generated for valid filters and reconcile with stored operational records.
------------------	---

Test Case	Input	Expected Output	Status
Fleet status report	Date: 26/08/2026; All vehicle categories	Report lists vehicles by Available, Reserved, On Lease, In Workshop and Out of Service.	Not Yet Executed
Maintenance due report	Due within next 7 days	Only vehicles meeting date/mileage criteria are displayed.	Not Yet Executed
Billing report	01/08/2026 to 31/08/2026	Invoice totals, paid and outstanding amounts match D5 records.	Not Yet Executed
Invalid date range	Date From after Date To	Error: Start date cannot be after end date.	Not Yet Executed
Export report	Click Export CSV/PDF after valid report generated	File generated containing displayed report data.	Not Yet Executed

4.0 CONCLUSION

The Analysis and Design of the Vehicle Services and Tracking System for NT Nigeria Limited provides a complete design blueprint for integrating the major stages of the company's vehicle leasing and fleet-service operations into a single web-based platform. The proposed NT Nigeria Limited VMS connects customer reservations, vehicle and chauffeur allocation, dispatch, telematics tracking, return inspection, maintenance scheduling and corporate billing through a centralized relational database.

The proposed system is structured around the operational areas already identified in the project:

- **Centralized Customer & Booking Management:** customer enquiries, quotations and reservation details are stored in one traceable booking record.
- **Real-Time Fleet Allocation:** vehicle availability is checked against active bookings and maintenance status before dispatch, reducing the possibility of conflicting assignments.
- **Integrated Vehicle Tracking:** GPS location, speed and odometer updates are linked directly to the active trip rather than existing as isolated tracking information.
- **Preventive Maintenance Control:** vehicle mileage, return inspection findings and service dates are used to trigger maintenance actions and prevent unavailable vehicles from being allocated.
- **Automated Billing & Invoicing:** approved lease rates and actual trip usage are combined to calculate base charges, excess mileage, extra days, driver-related charges and applicable penalties.
- **Management Reporting & Accountability:** fleet utilization, active trips, maintenance due and invoice/payment summaries can be generated from the same centralized data source.

The Required Logical DFD, data dictionary, decision tables, proposed ERD, database tables, screen layouts, form/report specifications, module specifications and test plans together describe how the proposed system should operate and how its major components interact. The remaining stakeholder survey responses from NT Nigeria Limited will be used mainly to validate the existing-system analysis

and to refine any company-specific business rules, approval procedures, rate structures and report requirements before final submission.

APPENDICES

Appendix A: Project Team Meeting Minutes

MINUTES OF PROJECT TEAM MEETING 1: Name Selection & Team Setup

Date: 12th July 2026 | Time: 10:00 PM – 11:00 PM

Venue: Google Meet

5 Present: 1. Orekoya Fiopefoluwa Opemipo - 240805003

2. Ogunlowo Omowunmi Ruth - 240805026

3. Obe Anthonia Morerinayo - 240805007

4. Abdulbaasit Isah - 240805050

5. Bakare Olaronke - 240805017

Agenda: Selection of case study, role distribution, and project scope definition.

Deliberations & Decisions: 1. The team reviewed transportation and fleet logistics operations in Nigeria, selecting NT Nigeria Limited's vehicle leasing operations as the core case study domain.

2. Group roles were assigned: Orekoya Fiopefoluwa Opemipo - 240805003 (Software Designer)

2. Ogunlowo Omowunmi Ruth - 240805026 (Project Leader)

3. Obe Anthonia Morerinayo - 240805007 (Data Model Designer)

4. Abdulbaasit Isah - 240805050 (Software Designer)

5. Bakare Olaronke - 240805017 (Secretary)

MINUTES OF PROJECT TEAM MEETING 2: Existing Workflow Analysis & Fact-Finding

Date: 22nd July 2026 | Time: 1:00 PM – 1:00 PM

Venue: Computer Science Frontage

Present: All team members present.

Agenda: Review of walk-in and WhatsApp intake workflows, vehicle categories, and bottleneck analysis.

Deliberations & Decisions: 1. Documented the company's multi-channel intake (reception paper slips and WhatsApp chats) and its multi-tier fleet (Toyota Coasters, Hiluxes, Prados, Corollas).

2. Identified critical bottlenecks: double-booking risks, detached GPS tracking dashboards, and delayed corporate billing reconciliation.

3. Software and Data Designers commenced drafting Current Physical and Logical DFDs.

Appendix B: SQL Schema Implementation DDL Code

The following PostgreSQL DDL implements the core relational structure specified in Section 3.3.2 for the proposed NT Nigeria Limited VMS.

```
-- NT Nigeria Limited VMS Core Relational Schema (PostgreSQL)
CREATE TABLE users (
  user_id SERIAL PRIMARY KEY,
  full_name VARCHAR(120) NOT NULL,
  username VARCHAR(80) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role VARCHAR(40) NOT NULL,
  user_status VARCHAR(20) DEFAULT 'Active'
);

CREATE TABLE customers (
  customer_id SERIAL PRIMARY KEY,
  customer_type VARCHAR(30) NOT NULL,
  customer_name VARCHAR(150) NOT NULL,
  phone_number VARCHAR(20) NOT NULL,
  email VARCHAR(120),
  address VARCHAR(250)
);

CREATE TABLE vehicles (
  vehicle_id SERIAL PRIMARY KEY,
  registration_no VARCHAR(20) UNIQUE NOT NULL,
  make VARCHAR(50) NOT NULL,
  model VARCHAR(80) NOT NULL,
  category VARCHAR(40) NOT NULL,
  current_mileage NUMERIC(10,1) DEFAULT 0,
  vehicle_status VARCHAR(30) NOT NULL,
  gps_device_id VARCHAR(80)
);

CREATE TABLE drivers (
  driver_id SERIAL PRIMARY KEY,
  driver_name VARCHAR(120) NOT NULL,
  phone_number VARCHAR(20) NOT NULL,
  license_no VARCHAR(50) UNIQUE NOT NULL,
  license_expiry DATE NOT NULL,
  availability_status VARCHAR(30) NOT NULL
);

CREATE TABLE rate_cards (
  rate_id SERIAL PRIMARY KEY,
  vehicle_category VARCHAR(40) NOT NULL,
  daily_rate NUMERIC(12,2) NOT NULL,
  mileage_allowance NUMERIC(10,1),
  excess_mileage_rate NUMERIC(12,2) DEFAULT 0,
  driver_allowance NUMERIC(12,2) DEFAULT 0,
  effective_date DATE NOT NULL
);
```

```

CREATE TABLE reservations (
    reservation_id SERIAL PRIMARY KEY,
    customer_id INT NOT NULL REFERENCES customers(customer_id),
    rate_id INT REFERENCES rate_cards(rate_id),
    created_by INT NOT NULL REFERENCES users(user_id),
    service_date TIMESTAMP NOT NULL,
    return_date TIMESTAMP NOT NULL,
    pickup_location VARCHAR(250) NOT NULL,
    vehicle_category VARCHAR(40) NOT NULL,
    booking_status VARCHAR(30) DEFAULT 'Pending',
    CHECK (return_date > service_date)
);

CREATE TABLE lease_contracts (
    contract_id SERIAL PRIMARY KEY,
    reservation_id INT UNIQUE NOT NULL REFERENCES reservations(reservation_id),
    contract_no VARCHAR(60) UNIQUE NOT NULL,
    base_lease_rate NUMERIC(12,2) NOT NULL,
    mileage_limit NUMERIC(10,1),
    terms TEXT,
    approval_status VARCHAR(30) DEFAULT 'Pending'
);

CREATE TABLE dispatches (
    dispatch_id SERIAL PRIMARY KEY,
    reservation_id INT UNIQUE NOT NULL REFERENCES reservations(reservation_id),
    vehicle_id INT NOT NULL REFERENCES vehicles(vehicle_id),
    driver_id INT REFERENCES drivers(driver_id),
    dispatch_datetime TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    initial_odometer NUMERIC(10,1) NOT NULL,
    dispatch_status VARCHAR(30) DEFAULT 'Released'
);

CREATE TABLE trips (
    trip_id SERIAL PRIMARY KEY,
    dispatch_id INT UNIQUE NOT NULL REFERENCES dispatches(dispatch_id),
    trip_start_time TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    trip_end_time TIMESTAMP,
    final_odometer NUMERIC(10,1),
    distance_covered NUMERIC(10,1),
    trip_status VARCHAR(30) DEFAULT 'Active'
);

CREATE TABLE tracking_logs (
    tracking_id BIGSERIAL PRIMARY KEY,
    trip_id INT NOT NULL REFERENCES trips(trip_id) ON DELETE CASCADE,
    latitude NUMERIC(10,7) NOT NULL,
    longitude NUMERIC(10,7) NOT NULL,
    speed NUMERIC(7,2),
    odometer NUMERIC(10,1),
    event_time TIMESTAMP NOT NULL,
    event_type VARCHAR(40)

```

);

```
CREATE TABLE maintenance_records (  
    maintenance_id SERIAL PRIMARY KEY,  
    vehicle_id INT NOT NULL REFERENCES vehicles(vehicle_id),  
    service_type VARCHAR(100) NOT NULL,  
    trigger_mileage NUMERIC(10,1),  
    due_date DATE,  
    fault_description TEXT,  
    maintenance_status VARCHAR(30) DEFAULT 'Open',  
    cost NUMERIC(12,2) DEFAULT 0  
);
```

```
CREATE TABLE invoices (  
    invoice_id SERIAL PRIMARY KEY,  
    reservation_id INT NOT NULL REFERENCES reservations(reservation_id),  
    base_amount NUMERIC(12,2) NOT NULL,  
    extra_mileage_charge NUMERIC(12,2) DEFAULT 0,  
    extra_day_charge NUMERIC(12,2) DEFAULT 0,  
    driver_charge NUMERIC(12,2) DEFAULT 0,  
    penalty_charge NUMERIC(12,2) DEFAULT 0,  
    total_amount NUMERIC(12,2) NOT NULL,  
    invoice_status VARCHAR(30) DEFAULT 'Draft'  
);
```

```
CREATE TABLE payments (  
    payment_id SERIAL PRIMARY KEY,  
    invoice_id INT NOT NULL REFERENCES invoices(invoice_id),  
    amount_paid NUMERIC(12,2) NOT NULL,  
    payment_method VARCHAR(30) NOT NULL,  
    payment_reference VARCHAR(100),  
    payment_date TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    payment_status VARCHAR(30) DEFAULT 'Confirmed'  
);
```

```
CREATE INDEX idx_reservation_dates  
    ON reservations(service_date, return_date, booking_status);  
CREATE INDEX idx_dispatch_vehicle ON dispatches(vehicle_id);  
CREATE INDEX idx_tracking_trip_time ON tracking_logs(trip_id, event_time DESC);  
CREATE INDEX idx_maintenance_vehicle_status  
    ON maintenance_records(vehicle_id, maintenance_status);
```

Appendix C: Backend Application Sample Logic

The sample Node.js/Express controller below illustrates the vehicle-allocation rule used by the Fleet Allocation & Dispatch Module. It checks booking confirmation, vehicle status, maintenance hold and overlapping allocations before committing the dispatch transaction.

```
// NT Nigeria Limited VMS - Fleet Allocation / Dispatch Controller
const pool = require('../config/database');

exports.createDispatch = async (req, res) => {
  const client = await pool.connect();
  const { reservation_id, vehicle_id, driver_id, initial_odometer } = req.body;

  try {
    await client.query('BEGIN');

    // 1. Lock and validate the reservation.
    const bookingResult = await client.query(`
      SELECT reservation_id, service_date, return_date,
             vehicle_category, booking_status
      FROM reservations
      WHERE reservation_id = $1
      FOR UPDATE
    `, [reservation_id]);

    if (bookingResult.rows.length === 0) {
      throw new Error('Reservation not found.');
```

```

}

// 3. Prevent allocation where an open maintenance record blocks the vehicle.
const maintenance = await client.query(`
  SELECT 1
  FROM maintenance_records
  WHERE vehicle_id = $1
        AND maintenance_status IN ('Open', 'In Progress', 'Hold')
  LIMIT 1
`, [vehicle_id]);

if (maintenance.rows.length > 0) {
  throw new Error('Vehicle has an active maintenance hold.');
```

```

}

// 4. Prevent overlapping use of the same vehicle.
const conflict = await client.query(`
  SELECT 1
  FROM dispatches d
  JOIN reservations r ON r.reservation_id = d.reservation_id
  WHERE d.vehicle_id = $1
        AND r.booking_status IN ('Confirmed', 'Active')
        AND r.service_date < $3
        AND r.return_date > $2
  LIMIT 1
`, [vehicle_id, booking.service_date, booking.return_date]);

if (conflict.rows.length > 0) {
  throw new Error('Vehicle is already allocated within the requested period.');
```

```

}

// 5. If a driver is supplied, confirm availability and licence validity.
if (driver_id) {
  const driverResult = await client.query(`
    SELECT driver_id, license_expiry, availability_status
    FROM drivers
    WHERE driver_id = $1
    FOR UPDATE
  `, [driver_id]);

  if (driverResult.rows.length === 0 ||
      driverResult.rows[0].availability_status !== 'Available' ||
      new Date(driverResult.rows[0].license_expiry) < new Date(booking.service_date)) {
    throw new Error('Selected driver is not eligible for this dispatch.');
```

```

  }
}

// 6. Create dispatch and active trip records atomically.
const dispatchResult = await client.query(`
  INSERT INTO dispatches
    (reservation_id, vehicle_id, driver_id, initial_odometer, dispatch_status)
  VALUES ($1, $2, $3, $4, 'Released')
  RETURNING dispatch_id

```

```

`, [reservation_id, vehicle_id, driver_id || null, initial_odometer]);

const dispatchId = dispatchResult.rows[0].dispatch_id;
await client.query(`
  INSERT INTO trips (dispatch_id, trip_status)
  VALUES ($1, 'Active')
`, [dispatchId]);

await client.query(
  `UPDATE vehicles SET vehicle_status = 'On Lease' WHERE vehicle_id = $1`,
  [vehicle_id]
);

if (driver_id) {
  await client.query(
    `UPDATE drivers SET availability_status = 'Assigned' WHERE driver_id = $1`,
    [driver_id]
  );
}

await client.query('COMMIT');
return res.status(201).json({
  success: true,
  dispatch_id: dispatchId,
  message: 'Vehicle allocation and dispatch completed successfully.'
});
} catch (error) {
  await client.query('ROLLBACK');
  return res.status(400).json({ success: false, error: error.message });
} finally {
  client.release();
}
};

```